ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(b)** 20 N	60 - 40.
3	**(a)** moves towards team A	Net force towards A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Weight equals air resistance.
5	**(a)** 200 N	600/3.
6	**(c)** magnetic force	Acts at a distance.
7	**(c)** A is true but R is false.	A is true; R is false because mass does not depend on gravity.
8	**(c)** 60 kg	Mass is unchanged.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R states exactly why friction is a contact force.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Mass is constant because the quantity of matter is the same; only weight changes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass remains 3 kg (1); Weight = 30 / 6 = 5 N (1).	1+1
12	Friction is lower (1); Less force opposes the sliding (1).	1+1
13	Spread the load (1); More grip and less pressure on the road (1).	1+1
14	It induces opposite charge on the near side (1); Attraction to nearer opposite charge is greater (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Friction = 30 N (1); With 20 N net force = 10 N backwards (1); The box slows down and stops (1).	1+1+1
16	Friction depends on the force pressing surfaces (1); Larger area but less pressure per unit area (1); These balance out (1).	1+1+1
17	Less distance to fall (1); Gravity acts for a shorter time (1); Speed has less time to increase (1).	1+1+1
	OR: Reduces the area facing the wind (1); Reduces air resistance (1); Helps ride faster (1).	1+1+1
18	Rub a steel needle in one direction with a magnet (1); It acquires magnetism (1); Test by picking pins (1).	1+1+1
	OR: Oil reduces friction between moving parts (1); No oil: great heating and wear (1); Engine may seize (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 60 - 40 = 20 N in the direction of the pull (1). (ii) Moving, since the net force is not zero (1). (iii) Opposite to the direction of motion (1). (iv) 60 - 10 = 50 N (1); rolling friction is much smaller than sliding friction, so less force is wasted (1).	1+1+1+2
20	Forces on the man: weight down, reaction up (1); Both equal to 700 N at rest (1); Constant speed: net force zero (1); Reading remains 700 N (1); Only acceleration changes reading (1).	1+1+1+1+1
	OR: Rubbing charges the comb (1); Charged comb attracts paper (1); Moist air has water vapour that conducts (1); Charge leaks away (1); No attraction (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that friction acts in the direction of motion; it acts **opposite** to the motion (or its tendency).
- Using kg as the unit of weight, or saying weight is constant everywhere.
- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.

- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.
- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 3 N to the right	Net 3 N to the left, so balance with 3 N right.
2	**(a)** both have the same sign of charge	Like charges repel.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Safety design.
4	**(c)** the ball is either negatively charged or neutral	Neutral bodies are also attracted.
5	**(c)** 400 N upward	Balanced.
6	**(c)** A is true but R is false.	A is true; R is false since paper has equal positive and negative charges.
7	**(a)** the ball has the same type of charge	Repulsion needs like charges.
8	**(b)** putting a heavy load on the box	Greater force pressing surfaces.
9	**(d)** A is false but R is true.	A is false: balanced forces cannot change the state of motion; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but the attraction is due to electrostatic force (charges), not gravity, so R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass is the amount of matter (1); Gravity acts on mass to give weight only (1).	1+1
12	No change in motion (1); Hand feels no extra force (1).	1+1
13	More charge gives greater divergence (1); Because repulsion between leaves is larger (1).	1+1
14	To remove static charge (1); Avoids sparks and fire (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Same charge: repel (1); Different sources can give opposite charges (1); Opposite charges attract (1).	1+1+1
16	Charge passes to disc, rod and leaves (1); Leaves get same charge (1); Repulsion makes them diverge (1).	1+1+1
17	Earthing equipment (1); Using anti-static materials (1); Keeping humidity adequate (1).	1+1+1
	OR: Tall objects attract discharge (1); Metal conducts charge to us (1); So we should avoid them (1).	1+1+1
18	Rubber is elastic and regains shape (1); Clay is plastic and keeps the changed shape (1); Force changes shape of both (1).	1+1+1
	OR: Gravitational force depends on masses (1); Masses are very small (1); Earth's gravity is much greater (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 40 x 10 = 400 N (1). (ii) Newton (1). (iii) Weight (1). (iv) Mass (5 kg) is the quantity of matter and stays the same everywhere; weight (50 N) is the gravitational pull of the Earth on the bag and depends on the place (2).	1+1+1+2
20	Gravitational: always attractive; all matter (1); Magnetic: attractive or repulsive; magnetic materials (1); Electrostatic: attractive or repulsive; charged bodies (1); Examples for each (1); All act at a distance (1).	1+1+1+1+1
	OR: Magnetise a needle (1); Push it through a cork (1); Float the cork in water (1); Needle aligns along N-S (1); Earth acts as a big magnet (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that friction acts in the direction of motion; it acts **opposite** to the motion (or its tendency).
- Using kg as the unit of weight, or saying weight is constant everywhere.
- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.
- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.
- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** friction at the axle and air resistance	Opposing forces.
2	**(d)** A is false but R is true.	A is false and R is true.
3	**(b)** in the same direction	Forces in the same direction add.
4	**(d)** a metal rod held in the bare hand	Charge on the metal rod flows through the hand to the earth.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(c)** there is no air resistance	No air.
7	**(a)** Weight depends on location but mass does not	Mass is constant.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(c)** A is true but R is false.	A is true; R is false because a spring balance measures force (weight), calibrated in newton.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Microscopic bumps on every surface cause friction.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The needle is a small magnet (1); It aligns with the Earth's magnetic field (1).	1+1
12	Contact area with the ground is less (1); Less friction with the road (1).	1+1
13	Surfaces still have microscopic irregularities (1); Fluid layers also give friction (1).	1+1
14	Skis need low friction to glide (1); Spikes need high friction for grip (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The upward 20 N supports part of the weight (1); First balance reads 30 N (1); The weight of the box is still 50 N (1).	1+1+1
16	Air resistance increases with speed (1); More engine force is needed (1); More fuel used (1).	1+1+1
17	Coin travels farther on a powdered board (1); Powder reduces friction (1); Less opposing force (1).	1+1+1
	OR: Moon's gravity is weaker (1); Less downward force (1); Same muscular force gives greater height (1).	1+1+1
18	Equal and opposite: net zero (1); Unequal: net in the direction of larger (1); Same direction: they add (1).	1+1+1
	OR: Water molecules have charged parts (1); Opposite charge attracts (1); Stream bends towards the comb (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Electrostatic force (1). (ii) Non-contact force (1). (iii) By rubbing (friction) it gains charge (1). (iv) Moist air is a conductor and carries the charge away, so the scale does not stay charged and cannot attract the paper (2).	1+1+1+2
20	Earth: 120 N (1); Moon: 20 N (1); Beam balance measures mass (same) (1); Spring balance measures weight (varies) (1); Mass 12 kg everywhere (1).	1+1+1+1+1
	OR: Gravity pulls skier down the slope (1); Friction and air resistance oppose (1); At the bottom friction continues (1); No driving force (1); So the skier slows and stops (1).	1+1+1+1+1

Mistakes to avoid:

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- Using kg as the unit of weight, or saying weight is constant everywhere.
- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.
- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.

- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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